Units and Dimension

1. THE NEED FOR MEASUREMENT

- **PHYSICS** - A quantitative science based on observing and measuring physical phenomena.
- **PHYSICAL QUANTITY** - Any quantity that can be measured (e.g. length, mass, time, force, temperature).
- **MEASUREMENT** - The process of comparing a physical quantity with a chosen, internationally accepted reference standard called a unit.
- **RESULT OF MEASUREMENT** - Expressed as a numerical value (n) and a unit (u).

Measurement = n x u

- **NEED FOR UNITS** - To ensure uniformity, reproducibility and meaningful comparison of measurements worldwide.

2. SYSTEM OF UNITS

- **HISTORICAL SYSTEMS**
  - CGS (or Gaussian): Centimetre (cm), Gram (g), Second (s)
  - FPS (BRITISH): Foot (ft), Pound (lb), Second (s)
  - MKS: Metre (m), Kilogram (kg), Second (s)

- **SI system (Systeme Internationale de unites)**

It is a rationalized M.K.S. system and modern, internationally accepted system.

- **Advantages** Coherent (derived units obtained by simple multiplication/division of base units without numerical factors. Metric

(uses power of 10), Widely accepted.

- **SI Base Quantities and Units -**

There are 7 fundamental quantities upon which the system is built.

Base Quantity	S.I. Base Unit	Symbol	Dimension
1) Length	metre	m	[L]
2) Mass	kilogram	kg	[M]
3) Time	second	s	[T]
4) Electric Current	ampere	A	[A] or [I]
5) Thermodynamic temperature	kelvin	K	[K] or [θ]
6) Luminous intensity	candela	cd	[cd]
7) Amount of substance	mole	mol	[mol]

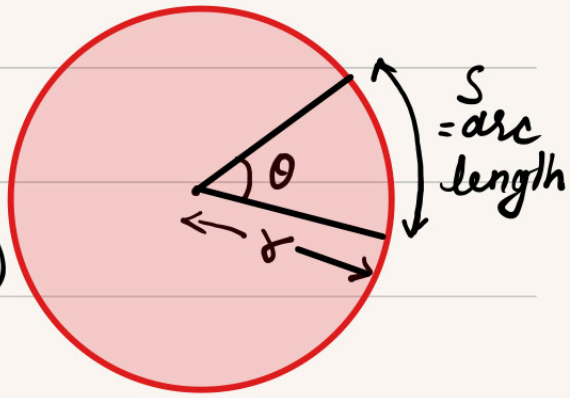
- **Two supplementary Quantities**

1) Plane Angle (θ)

↓ 2-D

S.I. unit – radian (rad)

other unit → degree (°)



$$\text{Plane angle} = \frac{\text{arc length}}{\text{radius}}$$

r → radius

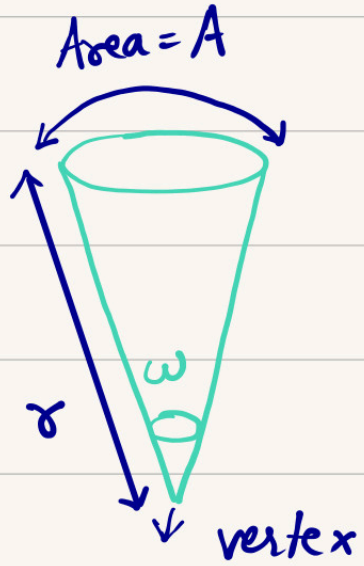
$180^\circ = \pi$ rad

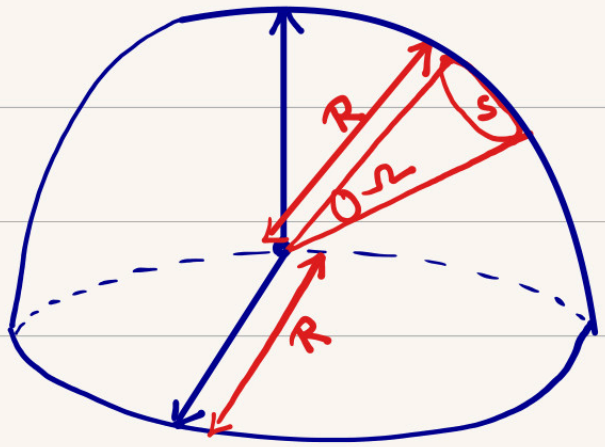
$360^\circ = 2\pi$ rad

2) Solid Angle (ω or Ω)

↓ 3-D

$$\omega = \Omega = \frac{\text{Area}}{(\text{radius})^2}$$





- **DERIVED QUANTITIES** physical quantities that can be expressed as combination of base quantities (e.g. of derived quantities, velocity (m/s), acceleration (m/s^2), force (N or $kg\text{-}m\text{-}s^{-2}$), Energy ($kg\text{-}m^2\text{-}s^{-2}$ or J))

- **CONVERSION OF UNITS**

$$n_1 u_1 = n_2 u_2$$

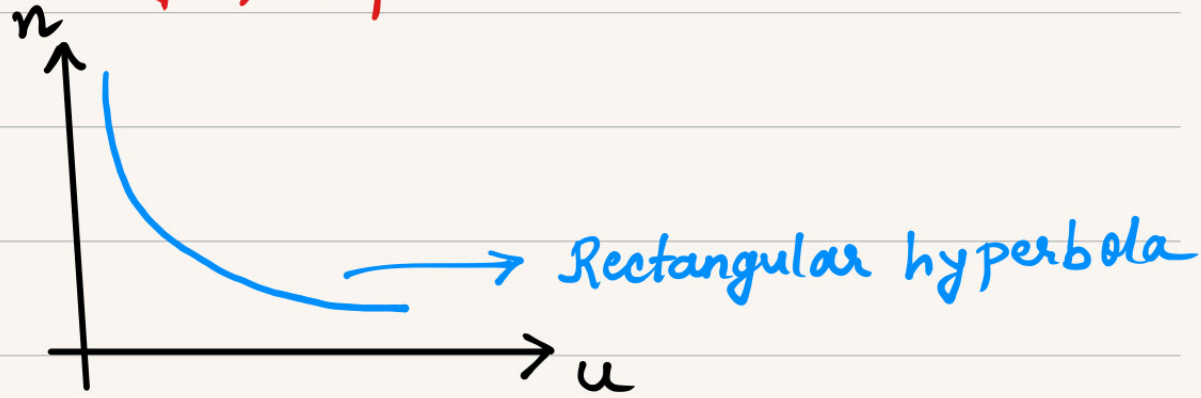
$$nu = \text{constant} \rightarrow xy = K \Rightarrow y = \frac{K}{x}$$

$$y \propto \frac{1}{x}$$

e.g. 5 (number (n)) km = 5000 m (unit (u))

Hence $n \uparrow \Rightarrow u \downarrow$

$n \downarrow \Rightarrow u \uparrow$



1) Speed or velocity

km/h → m/s

$n_1 u_1 = n_2 u_2$

$18 \frac{km}{h} = n_2 \times \frac{m}{s}$

$\frac{18 \times 1000 m}{3600 s} = n_2 \times \frac{m}{s}$

$n_2 = 18 \times \frac{5}{18} = 5 \frac{m}{s}$

FAMOUS CONVERSIONS

1) To convert km/h into m/s, multiply given speed by $\frac{5}{18}$.

e.g. 1) $36 \frac{km}{h} = 10 \frac{m}{s}$ 2) $90 km/h = 25 m/s$

and vice-versa

To convert $\frac{m}{s}$ in $\frac{km}{h}$, multiply by $\frac{18}{5}$.

e.g. 1) $20\frac{m}{s} = 72\frac{km}{h}$ 2) $35\frac{m}{s} = 126\frac{km}{h}$

2) Density $= \frac{\text{mass}}{\text{volume}}$

S.I. unit $\rightarrow \frac{kg}{m^3}$

e.g. $1000\frac{kg}{m^3} = 1\frac{g}{cm^3}$

e.g. Density of mercury is $13.600\frac{kg}{m^3}$ in M.K.S. and it will be $13.6\frac{g}{cm^3}$ in C.G.S.

- COMMON S.I. PREFIXES

These are used to denote multiples

and sub-multiple of units.

Prefix	Symbol	Factor	Prefix	Symbol	Factor
tera	T	10^{12}	deci	d	10^{-1}
giga	G	10^{9}	centi	c	10^{-2}
mega	M	10^{6}	milli	m	10^{-3}
kilo	k	10^{3}	micro	μ	10^{-6}
hecto	h	10^{2}	nano	n	10^{-9}
deca	da	10^{1}	pico	p	10^{-12}
			femto	f	10^{-15}

3. SIGNIFICANT FIGURES

Digits in a measured value that are known reliably plus the first uncertain digit. They indicate the precision of the instrument.

RULES FOR IDENTIFYING SIGNIFICANT FIGURES

1) All non zero digits are significant.

(e.g. 123.45 → 5 sig. figures)

2) Zeroes between two non-zero digits are significant (e.g. 1.002 → 4 sig. figures, 5.03 → 3 sig. figures)

3) Leading zeroes (Zeroes to the left of the first non zero digit) are not significant. They only indicate the position of the decimal point. (e.g. 0.0062 → 2 sig. fig., 0.051 → 2 sig. fig.).

4) Trailing Zeroes (Zeroes to the right of the last nonzero digit) in a number with a decimal point are significant. (e.g. 2.500 → 4 sig. fig., 0.06900 → 4 sig. fig.)

5) Trailing Zeroes in a number without a decimal point are not counted. (e.g. 4700 m → 2 sig. fig., 500000 → 1 sig. fig.

6) WHOLE NUMBERS (EXACT NUMBERS)

Numbers that are definitions or count of objects contain infinite significant figures. (e.g. 5 apples, 1 min = 60 sec have infinite significant figures.)

RULES FOR ROUNDING OFF

1) If the digit to be dropped is greater than 5, increase preceding digit by 1.

e.g. 2.746 → 2.75

(6 ↓ to drop)

2) If the digit to be dropped is less than 5, keep the preceding digit unchanged.

e.g. → 1.743 → 1.74

(3 ↓ to drop)

3) If the digit to be dropped is = 5 (followed by nothing/zero digits, then two possibilities may occur :

A) Increase the preceding digit by 1, if it is odd. (e.g. 2.735 round to 3 sig. figs → 2.74).

B) Keep the preceding digit unchanged if it is even. (e.g. 2.745 round to 3 sig.figs → 2.74).

4) RULES FOR ARITHMETIC OPERATIONS

1) Addition / Subtraction

The final result should have the same number of decimal places as the number in the calculation with the fewest decimal places. (e.g. 436.32 + 227.2 + 0.301 = 663.82 → Report as 663.8 → 1 decimal place).

2) Multiplication / Division

The result should have the

same number of significant figures as the in the calculation with fewest significant figures.
(e.g. $4.237/2.51 = 1.6880\ldots$ → Report as 1.69 → 3 sig fig.)

5) Scientific Notation

Expressing a number as $a \times 10^b$, where $1 < a < 10$. All digit in 'a' are significant
(e.g. 4700m with 4 sig. figs = 4.700×10^3 m)

6) Order of Magnitude

The nearest power of 10 in $x \times 10^p$, where $0.5 \leq x < 5$, is called order of magnitude.
(e.g. Diameter of Earth = 1.28×10^7 m, order of magnitude = 7)

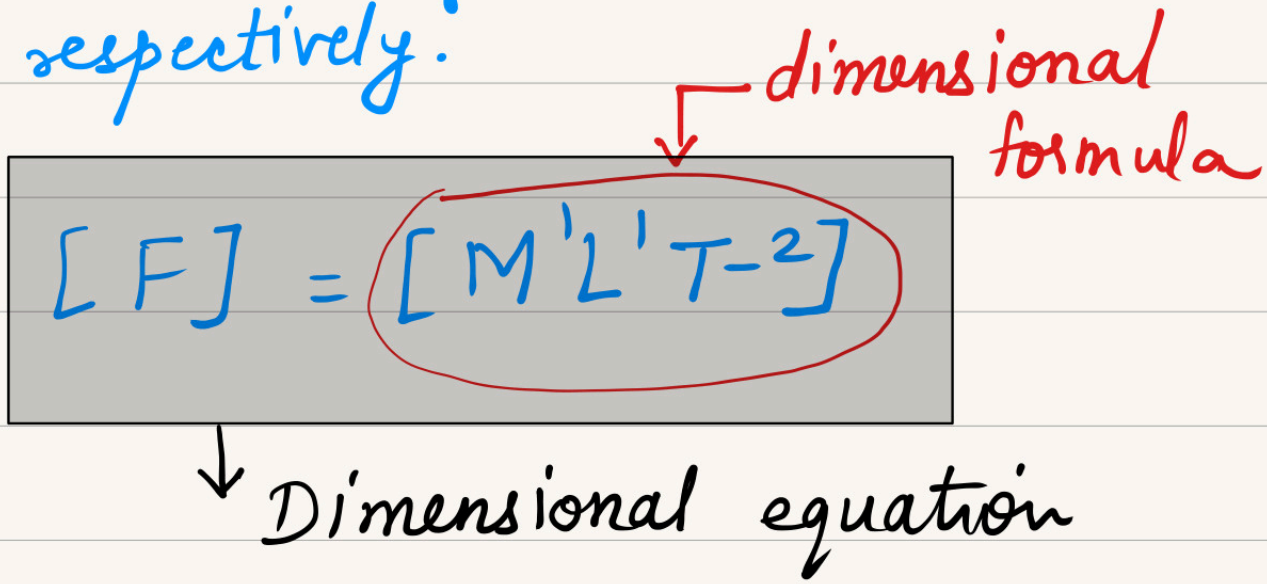
7) Dimensions

These are powers of fundamental

quantities in order to obtain unit of the physical quantity.

e.g. $[\text{force}] = [M^1L^1T^{-2}]$

Dimensions of M, L and T are 1, 1, -2 respectively.



* A typical dimensional formula in terms of all 7 fundamental quantities

$[M\ L\ T\ A\ K\ mol\ cd]$

* Density $= \frac{\text{mass}}{\text{volume}}$

$$[D] = [M^1 L^{-3} T^0]$$

Note

1) Any quantity which can have units must have dimensions [Except supplementary quantities]
2) Dimensionless quantities can have units (in plane angle and solid angle viz. supplementary quantities)
3) Following Quantities are dimensionless

- Plane angle
- Solid angle

- Refractive index
- Relative density
- Relative permeability
- Relative permittivity
- Strain
- Coefficient of friction
- All trigonometric functions, exponential functions, logarithm functions are dimensionless.

4) Constants may or may not have dimensions.

5) Following quantities have SAME DIMENSIONS

a) Planck's constant and angular momentum

b) Acceleration due to gravity and gravitational field intensity, acceleration.

c) Force, weight, thrust and energy gradient

d) Rydberg's constant, Propagation constant, Wave number

e) Boltzmann's constant, Gas constant, Entropy.

f) Momentum, impulse

g) Surface tension, force gradient

h) Gravitational potential, square of velocity, $[\mu_0 \varepsilon_0]^{-1}$ { where μ_0 is permeability of free space and ε_0 is permittivity of free space }.

i) $[\text{Energy}] = [M^1 L^2 T^{-2}]$

potential energy, kinetic energy, heat energy, work, torque, light energy etc.

Mechanical power, Electrical power, Thermal power, Emissive power.

- Field intensity

$[\text{Gravitation field intensity}] = [L^1T^{-2}]$

$[\text{Electric field intensity}] = [M^1L^1T^{-3}A^{-1}]$

$[\text{Magnetic field}] = [M^1T^{-2}A^{-1}]$

Dimensional formulae of physical quantities (Some examples from mechanics)

Although we may consider every physical quantity to expsess it as dimensional formulae, but

in the following table, some popular physical quantities are covered →

S.No.	Physical Quantity	Dim. formula	S.I. unit
1.	Velocity	$[M^0LT^{-1}]$	m/s
2.	Acceleration	$[M^0LT^{-2}]$	m/s^2
3.	Force	$[M^1L^1T^{-2}]$	N
4.	Work	$[M^1L^2T^{-2}]$	J
5.	Energy	$[M^1L^2T^{-2}]$	J
6.	Torque	$[M^1L^2T^{-2}]$	N-m
7.	Power	$[M^1L^2T^{-3}]$	W
8.	Momentum	$[M^1L^1T^{-1}]$	kg-m/s
9.	Impulse	$[M^1L^1T^{-1}]$	N-s
10.	Angle	$[M^0L^0T^0]$	radian
11.	Strain	$[M^0L^0T^0]$	No units
12.	Stress	$[M^1L^{-1}T^{-2}]$	N/m^2

For NEET purpose, student should cover following additional quantities→

1) Permittivity of free space (ε_0)

Coulomb's law (in electrostatics)

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$$

$$[\varepsilon_0] = \left[\frac{q_1 q_2}{F r^2}\right]$$

$$= \frac{[A^2T^2]}{[M^1L^1T^{-2}][L^2]} = [M^{-1}L^{-3}T^4A^2]$$

2) Permeability of free space (μ_0)

let,

speed of light in vacuum (c)

$$c^2 = \frac{1}{\mu_0 \varepsilon_0} \Rightarrow \mu_0 = \frac{1}{c^2 \varepsilon_0}$$

$$= \frac{1}{[L^2 T^{-2}] \times [M^1 L^{-3} T^4 A^2]}$$

$$[\mu_0] = [M^1 L^1 T^{-2} A^{-2}]$$

3) Coefficient of viscosity (η)

Viscous force $[F] = [6\pi\eta r v]$

$$[\eta] = \frac{[F]}{[r][v]}$$

$$= \frac{[M^1 L^1 T^{-2}]}{[L^1][L^1 T^{-1}]}$$

$$[\eta] = [M^1 L^{-1} T^{-1}]$$

4) Latent heat [L]

Heat energy (Q) = mass (m) × L

$$L = \frac{Q}{m}$$

$$[L] = \frac{[M^1 L^2 T^{-2}]}{[M^1]}$$

$$[L] = [M^0 L^2 T^{-2}]$$

5) specific heat (c)

Heat energy (Q) = $m\, c\, \Delta\theta$

$$c = \frac{Q}{m \Delta\theta}$$

$$[c] = \frac{[M^1 L^2 T^{-2}]}{[M^1 K^1]}$$

$$[c] = [M^0 L^2 T^{-2} K^{-1}]$$

6) Thermal current

$$i = \frac{dQ}{dt} = \frac{K A \Delta\theta}{L} = \frac{\text{Energy}}{\text{time}}$$

where, K is coefficient of thermal conductivity

$$[i] = [M^1 L^2 T^{-3}]$$

7) Coefficient of thermal conductivity (K)

$$K = \frac{iL}{A\Delta\theta}$$

$$[K] = \frac{[M^1 L^2 T^{-3}][L^1]}{[L^2][K^1]}$$

$$[K] = [M^1 L^1 T^{-3} K^{-1}]$$

8) Thermal resistance

$$R_{th} = \frac{L}{KA}$$

$$[R_{th}] = \frac{[L]}{[M^1L^1T^{-3}K^{-1}][L^2]}$$

$$= [M^1L^{-2}T^3K^1]$$

9) Electric potential (V)

$$V = E\gamma = \frac{F}{q}\cdot\gamma = \frac{F}{It}\cdot\gamma$$

$$[V] = \frac{[M^1L^1T^{-2} \times L^1]}{[A^1T^1]}$$

$$[V] = [M^1L^2T^{-3}A^{-1}]$$

10) Resistance

$$V = IR \Rightarrow R = \frac{V}{I}$$

$$[R] = \frac{[M^1L^2T^{-3}A^{-1}]}{[A^1]}$$

$$[R] = [M^1L^2T^{-3}A^{-2}]$$

11) Resistivity or Specific Resistance (ρ)

$$\rho = \frac{RA}{L} = \frac{[M^1L^2T^{-3}A^{-2}][L^2]}{[L^1]}$$

$$[\rho] = [M^1L^3T^{-3}A^{-2}]$$

12) Conductivity (σ)

$$[\sigma] = \frac{1}{[\rho]} = \frac{1}{[M^1L^3T^{-3}A^{-2}]}$$

$$[\sigma] = [M^{-1}L^{-3}T^3A^2]$$

13) Energy density

$$\frac{[E]}{[V]} = \frac{[M^1L^2T^{-2}]}{[L^3]} = [M^1L^{-1}T^{-2}]$$

APPLICATIONS OF DIMENSIONAL ANALYSIS

1) Conversion of units of a physical quantity from one system to another

2) To check accuracy of a formula or an equation

3) Derivation of formula

Explanation of Applications →

1) Conversion of units

Magnitude must be same.

$$n_1 u_1 = n_2 u_2 \Rightarrow n \propto \frac{1}{u}$$

1 m = 100 cm

↓ Large unit

↓ Small magnitude

} and vice-versa

steps to solve

Write units of physical quantity on both sides and write

units according to system of units. Then apply basic conversions like 1m = 100cm ; 1kg = 1000g etc.

$$n_2 = n_1\left[\frac{M_1}{M_2}\right]^a\left[\frac{L_1}{L_2}\right]^b\left[\frac{T_1}{T_2}\right]^c$$

2) Check the accuracy of formula

(i) Dimensionally correct equation may or may not be physically correct.

(ii) Dimensionally incorrect equation is always physically incorrect.

$S_{nth} = u + \frac{a}{2}(2n-1)$, is dimensionally incorrect but we treat it as physically correct.

iii) For the accuracy, we need principle of homogeneity.

PRINCIPLE OF HOMOGENEITY

dimensions of LHS = dimensions of RHS

Two quantities can be added or subtracted, only when their dimensions are same.

- Note: Division / Product of quantities having different dimensions are allowed.
- e^{AB}, $e^{A/B}$, $\ln\left(\frac{A}{B}\right)$, $\sin(AB)$ must be dimensionless.
- e^{A+B}, $\sin(A+B)$ → A and B both must be dimensionless.

e.g. $V = V_0 e^{-Kt^2}$

Kt^2 should be dimensionless.

$[K] = [T^{-2}]$

$\therefore [K] = [M^0 L^0 T^{-2}]$

3) Derivation of formula

e.g. Centripetal force acting on a mass (m) in circular path depends on velocity (v) and radius (r). Derive an expression for centripetal force.

LIMITATIONS

i) We can not derive formula in which log, exponential, trigonometric functions are involved.

ii) We can not derive formula in which addition and subtraction of two or more terms are involved.

iii) We can derive formula in which product and division of physical quantities are involved but we can not get any information about the dimensionless constants involved.

iv) It doesnot provide any info about

scalars or vectors.

e.g. 1) Derive an expression for the time period of a simple pendulum, which depends on length (l) and acceleration due to gravity (g).

solution $T \propto l^a g^b$ (step 1)

now, write dimensional formula of T, l, g and equate powers to get values of 'a' and 'b' to derive exp. for T.

2) In a unit system force, velocity, time are taken as fundamental quantities, find the dimensional formula for mass in this system.

solution let mass $\propto F^a V^b T^c$

Now write dim. formula for F, V and T and equate powers to get values of a, b and c to derive exp for mass in terms of F, V & T.

ERROR ANALYSIS

It is difference between true value and measured value of a physical quantity.

ERROR = TRUE VALUE − MEASURED VALUE

Let a student / person takes n observations of a physical quantity 'a' as $a_1, a_2, a_3, \ldots, a_n$

mean value $(a_m) = \dfrac{a_1 + a_2 + \ldots + a_n}{n}$

- True value (a_t)

If not given,

(a_m) mean value = True value (a_t)

$$a_m = a_t$$

1) Absolute error (Δa)

In every observation,

$$\Delta a_1 = a_m - a_1$$

$$\Delta a_2 = a_m - a_2$$

$$\vdots \qquad \vdots \qquad \vdots$$

$$\Delta a_n = a_m - a_n$$

2) Mean absolute error

- Average of all absolute errors
- Defined for final reading

$$\rightarrow \quad \Delta a_m = \frac{|\Delta a_1| + |\Delta a_2| + \ldots + |\Delta a_n|}{n}$$

- Quantity (A)

Representation of a physical quantity A, within error limits is $A \pm \Delta A$.

3) Fractional Error

$$\frac{\Delta a}{a} \text{ (or) } \frac{\Delta a_m}{a_m}$$

4) Percentage Error

$$= \frac{\Delta a}{a} \times 100$$

% error = fractional error x 100

- Change = Final value − Initial value

→ If change is positive, then gain or increase is shown by quantity.

→ If change is negative, then loss or decrease is shown by quantity.

- Fractional change in error $\left(\frac{\Delta x}{x}\right)$

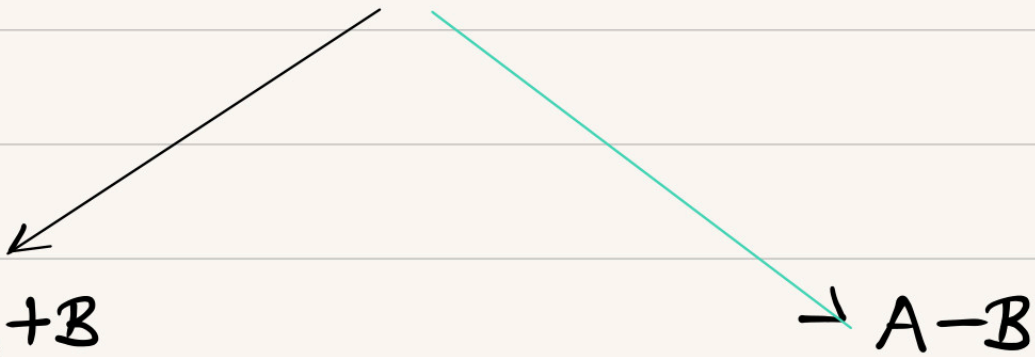
(i) for smaller change

Use differentiation

ERRORS IN MEASUREMENT

1) Error in addition / subtraction

$$x = A \pm B$$



Errors always add

$$\Delta x = \Delta A + \Delta B$$

2) Error in Product / Division

$$x = AB \quad (\text{or}) \quad x = \frac{A}{B}$$

$$\frac{\Delta x}{x} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$$

In product / division, fractional error in result is equal to the sum of fractional errors in quantities involved.

3) $$\frac{1}{x} = \frac{1}{A} + \frac{1}{B} \qquad \text{——(1)}$$

e.g. $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$; $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$ etc.

Eq^n (1) can be re-written as,

$$x^{-1} = A^{-1} + B^{-1}$$

differentiation w.r.t. variables

$$\frac{\Delta x}{x^2} = \frac{\Delta A}{A^2} + \frac{\Delta B}{B^2}$$

c) $y = x^n$

$$\frac{\Delta y}{y} = n \times \frac{\Delta x}{x}$$

d) $y = \frac{x^n z^m}{A^p}$

$$\frac{\Delta y}{y} = n \times \frac{\Delta x}{x} + m \times \frac{\Delta z}{z} + p \times \frac{\Delta A}{A}$$

e) $y = \sin x$

$$\frac{\Delta y}{y} = \cos x \times \frac{\Delta x}{x}$$

f) $y = e^x$

$$\frac{dy}{dx} = e^x \times \frac{\Delta x}{x}$$

(ii) Bigger Change

If change is more than 5 %.

then $\frac{\Delta x}{x} = \frac{x_f - x_i}{x_i}$

LEAST COUNT

- It is defined for particular instrument.
- It is the smallest value of a physical quantity, which can be measured accurately with an instrument.

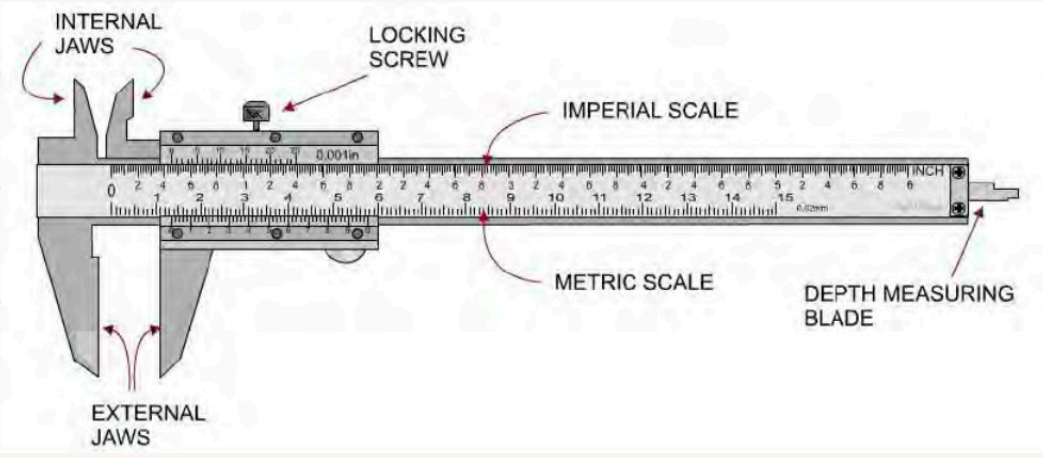
Two instruments are discussed →

1) Vernier calipers
2) Screw Gauge

1) Vernier Calipers

Two scales are considered

a) main scale



M.S.D. → value of 1 main scale division

V.S.D. → A division on vernier scale, which exactly coincides with a division on main scale.

$$a \text{ V.S.D.} = b \text{ M.S.D.}$$

$$1 \text{ V.S.D.} = \frac{b}{a} \text{ M.S.D.} \quad \text{—(1)}$$

$$\text{Least count (L.C.)} = 1 \text{ M.S.D.} - 1 \text{ V.S.D.}$$

$$= 1 \text{ M.S.D.} - \frac{b}{a} \text{ M.S.D.}$$

(The value of M.S.D. will be given in question itself)

$$\text{Hence, L.C.} = \left(\frac{a-b}{a}\right) \times \text{M.S.D.}$$

Total scale reading = (main scale reading + vernier scale reading)

- Zero error in vernier calipers

When jaws are closed and zero mark of main scale don't coincide

with zero of vernier scale.

There are 2 types of zero error

positive (where vernier zero is to the right of main scale zero).

negative (where vernier zero is to the left of main scale zero).

Zero error value is subtracted and added respectively from Total value in case of positive and negative zero error.

e.g. In vernier calipers, 1 division of main scale is 0.1 cm. If 'n' parts of main scale coincides with (n+2) parts of vernier calipers, then find least count.

solution $(n+2)$ V.S.D. $= n$ M.S.D.

$$1 \text{ V.S.D.} = \frac{n}{n+2} \text{ M.S.D.}$$

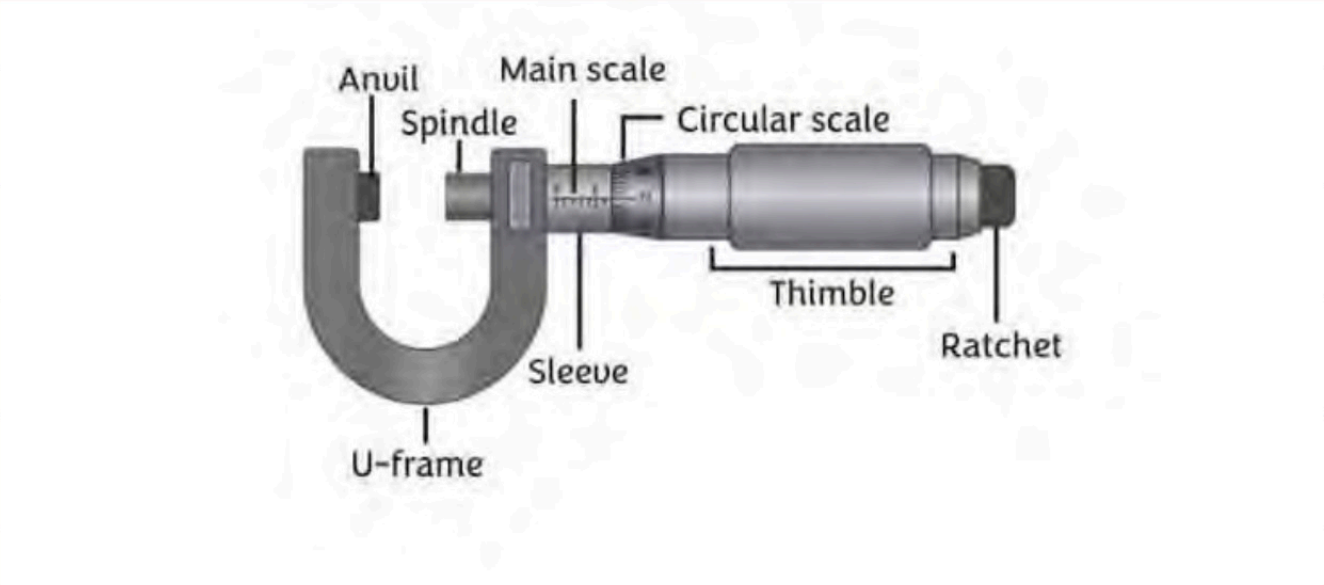
$$\text{L.C.} = 1 \text{ M.S.D.} - 1 \text{V.S.D} = 1\text{M.S.D} - \left(\frac{n}{n+2}\right)\text{M.S.D}$$

$$L.C. = \frac{2}{n+2} \quad M.S.D. = \frac{2}{n+2} \times 0.1\,cm = \frac{1(n+2)}{5}\,cm$$

2) SCREW GAUGE

Two scales are considered.

a) Main scale b) Circular scale



PITCH It is the linear distance covered in 1 rotation.

$$pitch = \frac{\text{Total linear distance covered on linear scale}}{\text{Number of rotations}}$$

$$\text{Least Count} = \frac{\text{Pitch}}{(\text{No. of divisions on circular Scale})}$$

- Total scale reading = (main scale reading + circular scale reading)

(e.g. diameter of a thin wire etc)

Zero Error in Screw Gauge

When the anvil and spindle are in contact and zero mark on the circular scale (thimble) does not coincide with the zero mark on the main scale, then zero error occurs.

Zero error is of 2 types:

- Positive zero error The thimble scale zero mark advances beyond main scale zero.
- Negative zero error The thimble scale zero mark is left behind the main scale zero mark.

Correction for zero error

Zero error value is subtracted and added respectively from Total value in case of positive and negative zero error.

Q) If a screw gauge has a positive zero error of 0.03 mm and the measured reading is 5.29 mm, find the corrected reading.

solution Corrected reading = (5.29 − 0.03) mm

As positive zero error is subtracted from measured reading.

ACCURACY and PRECISION IN MEASUREMENT

Accuracy tells how close a given set of measurements (observations or readings) are to their true value.

Precision is how close the measure-ments are to each other.

e.g. If the mass measure of a body